Electric Potential and Potential Energy

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Electric Potential and Potential Energy

Electrostatic Potential Energy

Change in potential energy of a system is defined as negative of the work done by conservative force in assembling the system from initial to final configuration.

Since, $dU = - dW_C$

Therefore, $\int_i^f dU = -\int_i^f \vec{F}_e . d\vec{r}$

$$(U_f - U_i) = -\int_i^f \vec{F}_e . d\vec{r}$$

Potential Energy of a system with two charges

It is defined as the work done against the electrostatic force between two charged particles in bringing them together at a particular separation from infinity.



$$dW_e = \vec{F}_e . d\vec{r}$$

$$dU = -\vec{F}_e . d\vec{r}$$

$$dU = -\frac{Kq_1 q_2}{r^2}(dr)\cos 0°$$

$$\int_i^f dU = -K q_1 q_2 \int_{r_i}^{r_f} \frac{dr}{r^2}$$

$$U_f - U_i = -K q_1 q_2 \left[-\frac{1}{r}\right]_{r_i}^{r_f}$$

$$U_f - U_i = Kq_1 q_2 \left[\frac{1}{r_f} - \frac{1}{r_i}\right]$$

Take r_i at infinity as a reference point and assign $U_\infty = U_i = 0$, also take $r_f = r$ and $U_f = U$. Then,

$$U - U_\infty = Kq_1 q_2 \left[\frac{1}{r} - \frac{1}{\infty}\right]$$

$$U = \frac{K q_1 q_2}{r}$$

This potential energy is also the work done by an external force in slowly bringing together two charged particles from infinity to a particular separation.

In other words,

$$dU = \left(dW_{ext}\right)_{slowly} = - dW_C$$

If q_1 and q_2 are of same sign, then U will be positive, i.e., positive external work has to be done in bringing the charges from infinity to some finite distance.

Similarly, if q_1 and q_2 are of opposite sign, then they will attract each other and U will be

negative, i.e., external agent will gain work in bringing the charges from infinity to some finite distance.

Electrostatic potential energy of a system of n-point charges

$$U_{sys} = \sum_{i=1}^{n} U_{ij} = \sum \frac{Kq_iq_j}{r_{ij}}$$

where U_{ij} = Potential energy (P.E.) of the pair q_i and q_j .
r_{ij} = Distance between r_i and r_j.

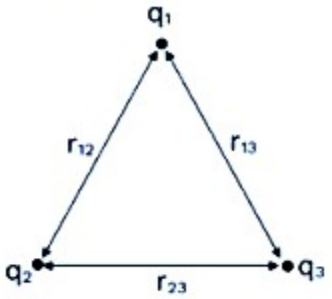
Total number of terms = $\frac{n(n-1)}{2} = {}^nC_2$

Q. **When the separation between two charges is increased, the electric potential energy of the charges**

(A) increases **(B) decreases**
(C) remain the same **(D) may increase or decrease**

Sol: Option (D) will be correct as the product q_1q_2 can be positive or negative.

Q. **Find the potential energy of given system.**



Sol: Here the potential energy of the system means the total external work done that needs to be done in order to move these charges slowly from infinite separation to the given separation.

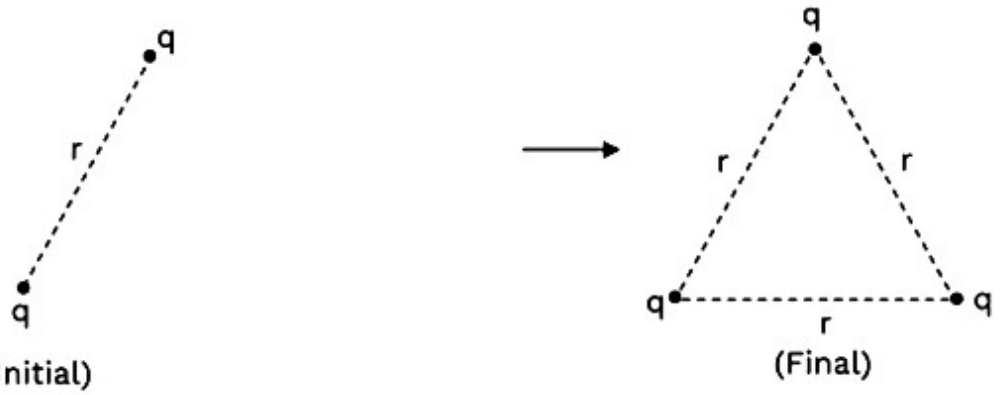
$$\therefore \quad W_{ext} = U_{sys} = U_{12} + U_{23} + U_{13}$$

$$U_{sys} = \frac{Kq_1q_2}{r_{12}} + \frac{Kq_2q_3}{r_{23}} + \frac{Kq_1q_3}{r_{13}}$$

Q. **Two identical charges are placed at the two corners of an equilateral triangle. The potential energy of the system is U. The work done in bringing an identical charge from infinity to the third vertex is**

(A) U **(B) 2U** **(C) 3U** **(D) 4U**

Sol: **(B)**

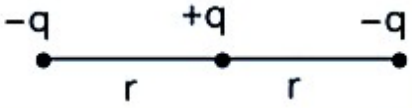


Initial potential energy, $U_i = \frac{Kq^2}{r} = U$

Final potential energy, $U_f = \frac{3Kq^2}{r} = 3U$

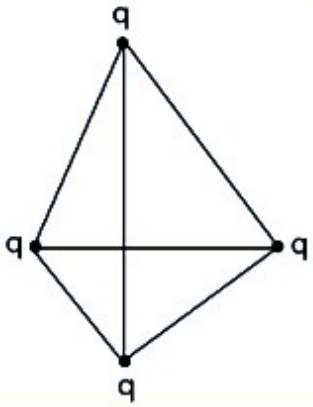
Then, $W_{ext} = U_f - U_i = 3U - U = 2U$

Q. **Find the electrostatic potential energy of the system of three points charge −q, +q and −q kept along a straight line as shown.**



Sol: $U_{sys} = \frac{-Kq^2}{r} - \frac{Kq^2}{r} + \frac{Kq^2}{2r} = \frac{-3}{2}\frac{Kq^2}{r} = \frac{-3q^2}{8\pi\varepsilon_0 r}$

Q. **Find the interaction energy of 4 point charges kept at vertices of the tetrahedron with edge length "a".**



Sol: All charges and distances are similar, so every pair has a potential energy,

$$U = \frac{q^2}{4\pi\varepsilon_0 a}$$

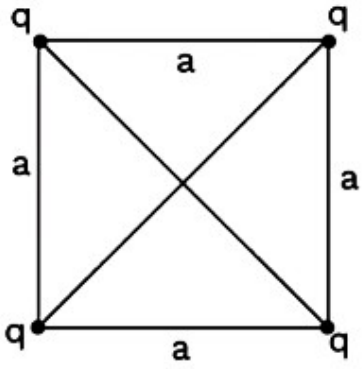
$$\text{Total pairs} = {}^4C_2 = \frac{4(4-1)}{2} = 6$$

$$\therefore \text{ Potential energy of the system, } U_{sys} = \frac{6q^2}{4\pi\varepsilon_0 a}$$

Q. **Four equal charges of charge q are placed at corner of a square of side a. Potential energy of the whole system is**

Sol: $$\text{Total pairs, } {}^4C_2 = \frac{4\times 3}{2} = 6$$

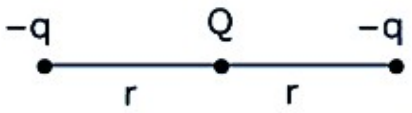
Out of these 6 pairs, 4 pairs are for the edges and 2 pairs
are for the diagonal lengths.



$$U_{sys} = 4\left(\frac{Kq^2}{a}\right) + 2\left(\frac{Kq^2}{\sqrt{2}\,a}\right)$$

$$U_{sys} = \frac{4Kq^2}{a}\left(1 + \frac{1}{2\sqrt{2}}\right)$$

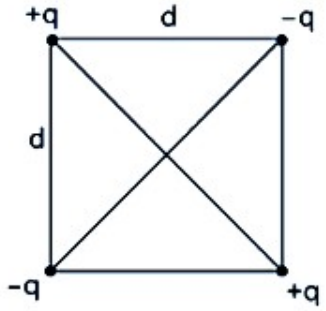
Q. **Three charges are placed as shown in figure if the electric potential energy of the system is zero, then the ratio Q : q is?**



Sol: Potential energy of the system, $U_{sys} = 2\left(-\frac{KqQ}{r}\right) + \frac{Kq^2}{2r} = 0$

Then, $\frac{Q}{q} = \frac{1}{4}$.

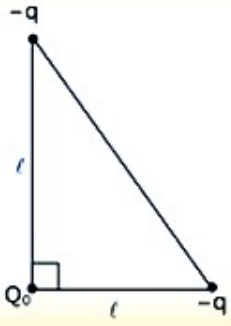
Q. **Calculate the minimum energy required to separate the charges to infinity kept at corners of a square shown in the given figure.**



Sol: Energy needed to separate all the charges will be equal to the negative of the potential energy of the system.

$$\therefore\ W_{ext} = -\left[\frac{-4q^2}{4\pi\varepsilon_0 d} + \frac{2q^2}{4\pi\varepsilon_0\sqrt{2}d}\right] = \frac{q^2}{2\pi\varepsilon_0 d}\left[2 - \frac{1}{\sqrt{2}}\right]$$

Q. **3 charges Q_0, $-q$ and $-q$ kept at the vertices of an isosceles right angle triangle are shown in the figure. The net electrostatic potential energy is zero if Q_0 is equal to**

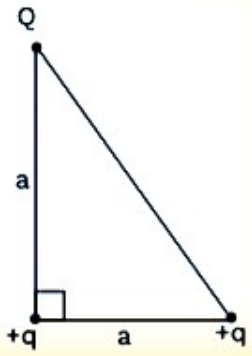


Sol: Potential energy of the system, $U = \frac{-2KqQ_0}{\ell} + \frac{Kq^2}{\sqrt{2}\ell} = 0$

$$\Rightarrow\ 2Q_0 = \frac{q}{\sqrt{2}}$$

$$\Rightarrow\ Q_0 = \frac{q}{2\sqrt{2}}$$

Q. **Three charges are placed as shown in figure. The net electrostatic energy of the configuration is zero if Q is equal to**

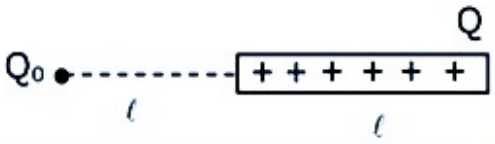


Sol: Potential energy of the system, $U = \frac{KQq}{a} + \frac{KQq}{\sqrt{2}a} + \frac{Kq^2}{a} = 0$

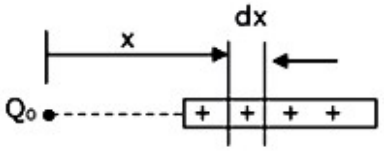
$$\Rightarrow Q\left(1 + \frac{1}{\sqrt{2}}\right) = -q$$

$$\Rightarrow Q = \frac{-\sqrt{2}\,q}{\sqrt{2}+1} \Rightarrow Q = \frac{-2q}{2+\sqrt{2}}$$

Q. **Calculate the interaction potential energy between a point charge Q_0 and an insulating rod with uniform charge Q distributed along its length l.**



Sol: Elemental charge on the rod, $dQ = \frac{Q}{\ell}dx$



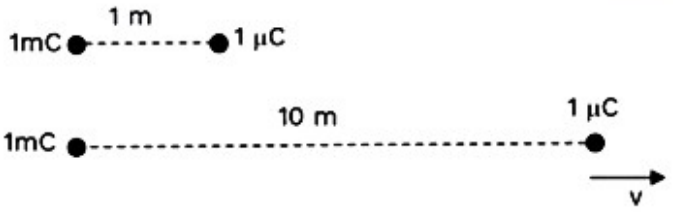
$$dU = \frac{KQ_0dQ}{x} = \frac{KQ_0Q}{\ell}\frac{dx}{x}$$

Total potential energy, $U = \int dU = \frac{KQ_0Q}{\ell}\int_{\ell}^{2\ell}\frac{dx}{x} = \frac{1}{4\pi\varepsilon_0}.\frac{Q_0Q}{\ell}\ln 2$

Q. **A particle of mass 2g and charge 1 μC is held at a distance of 1 m from a fixed charge of 1 mC. If the particle is released, it will be repelled. When the separation between the particles becomes 10 m, then what is the speed of the movable charge?**

Sol: Initial potential energy,

$$U_i = \frac{9\times10^9\times\left(10^{-3}\right)\left(10^{-6}\right)}{1} = 9\,J$$



Initial kinetic energy, $K_i = 0$.

At the instant when the separation between the charges is 10 m,

Final potential energy, $U_f = \frac{9\times10^9\times\left(10^{-3}\right)\left(10^{-6}\right)}{10} = 0.9\,J$

If the velocity of the movable charge is v, then,

Final kinetic energy, $K_f = \frac{1}{2}\left(2\times10^{-3}\right)v^2 = 10^{-3}v^2$

The total mechanical energy of the system remains constant.

So, $U_f + K_f = U_i + K_i$

$\Rightarrow 0.9 + 10^{-3}v^2 = 9 + 0 \Rightarrow v^2 = 81 \times 10^2 \Rightarrow v = 90$ m/s

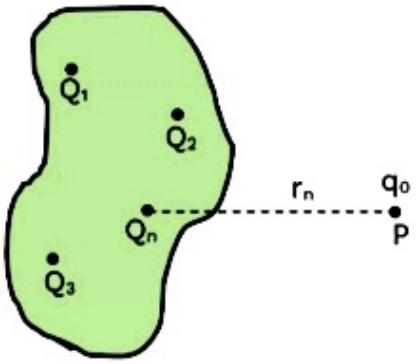
Electrostatic Potential

Work done by an external force in bringing a unit positive charge from infinity to a point is equal to electrostatic potential (V) at that point.

If U is the electrostatic potential energy of interaction of a system of charges with 'q_0' at a particular point P,

Then, electric potential at P is given by:

$$V = \lim_{q_0 \to 0}\left(\frac{U}{q_0}\right)$$

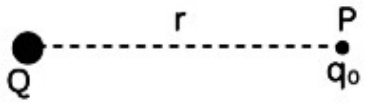


System of charges

Here V is independent of the test charge q_0 and electric potential is a scalar quantity and it's unit is volt (V) or J/C with dimension $M^1L^2T^{-3}A^{-1}$.

Electric potential due to a point charge

Electric potential due to a point charge Q at a distance r i



$$V = \frac{U}{q_0} = \frac{1}{q_0}\left[\frac{KQq_0}{r}\right]$$

$$\Rightarrow \quad V = \frac{KQ}{r}$$

Q. **Electric field and electric potential due to a point charge at a point, are 500 V/m and 3000V respectively. What is the distance between the point charge and the point?**

Sol: Electric field due to point charge, $E = \frac{KQ}{r^2} = 500$

And, potential due to point charge, $V = \frac{KQ}{r} = 3000$

So, $r = \frac{V}{E} = \frac{3000}{500} \Rightarrow r = 6$ m.

Q. **Three equal charges are placed at the three corners of an equilateral triangle.**

Will E and V exist at the centre of the triangle?

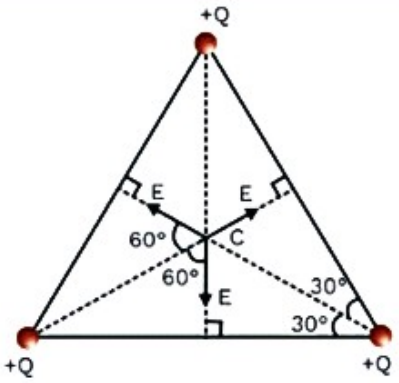
Sol: Due to symmetry,

Electric field at the centre, E = 0.

But, the electric potential will be zero.

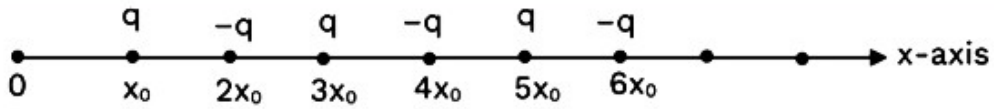
Electric potential at centre,

$$V = \frac{3KQ}{\left(\frac{a}{\sqrt{3}}\right)} = \frac{3\sqrt{3}KQ}{a} \quad \text{or } V \neq 0.$$



Q. **A charge +q is fixed at each of the point $x = x_0, 3x_0, 5x_0, \ldots\ldots\infty$ on the x-axis and a charge −q is fixed at each of the point $x = 2x_0, 4x_0, 6x_0, \ldots\ldots\infty$. If x_0 is a positive constant, find the potential at origin.**

Sol:



Electric potential at the origin,

$$V = \frac{Kq}{x_0} - \frac{Kq}{2x_0} + \frac{Kq}{3x_0} \ldots\ldots = \frac{Kq}{x_0}\left[1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \ldots\ldots\right] = \frac{Kq\ln 2}{x_0}$$

Electric potential due to a uniformly charged ring

(a) Potential at the centre of the ring

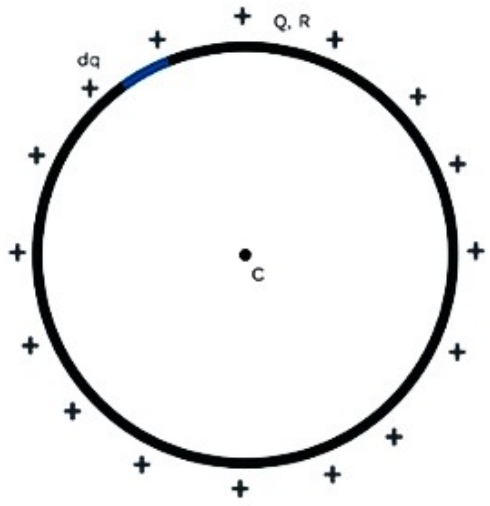
Consider an elemental charge dq on the periphery of the ring.

Potential at centre C due to elemental charge dq,

$$dV = \frac{Kdq}{R}$$

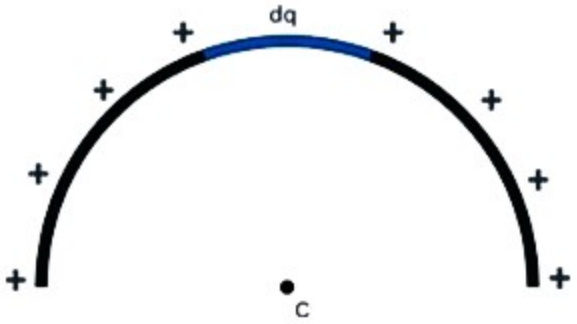
Total potential, $V = \int dV = \int \frac{Kdq}{R}$

$$V = \frac{K}{R}\int dq = \frac{KQ}{R}$$



Q. **A half ring of radius R has a charge of λ per unit length. The potential at the centre of the half ring is**

Sol: Charge on a small element, $dq = \lambda\, dx$



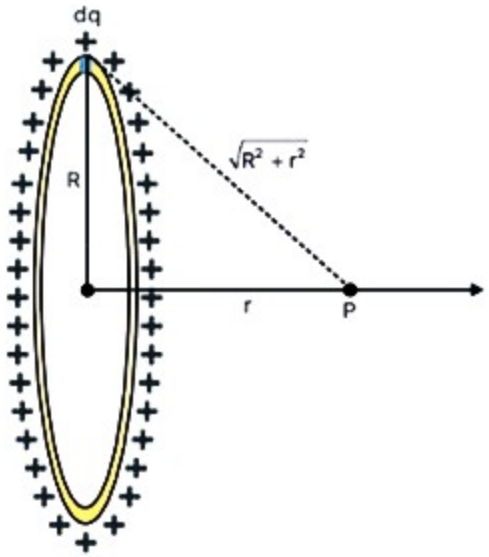
$$\therefore\ dV_c = \frac{K\,dq}{R} = \frac{K\lambda}{R}dx$$

$$\text{And, } V_c = \frac{K\lambda}{R}\int dx \Rightarrow V_c = \frac{K\lambda}{R}(\pi R)$$

$$\text{So, } V_c = K\lambda\pi$$

(b) Potential at a point on the axis of the ring

For a point P lying on the axis of the ring at a distance r from its centre, electric potential due to an elemental charge dq is,



$$dV_p = \frac{K\,dq}{\sqrt{R^2 + r^2}}$$

$$\text{Total potential, } V_p = \frac{K}{\sqrt{R^2 + r^2}}\int dq$$

$$V_p = \frac{KQ}{\sqrt{R^2 + r^2}}$$

(i) At centre of the ring (r = 0),

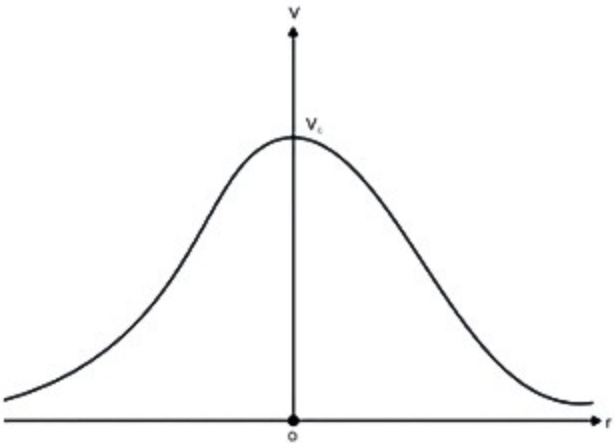
$$V_{centre} = \frac{KQ}{R}$$

(ii) At a point which is very far from centre $(r >> R)$,

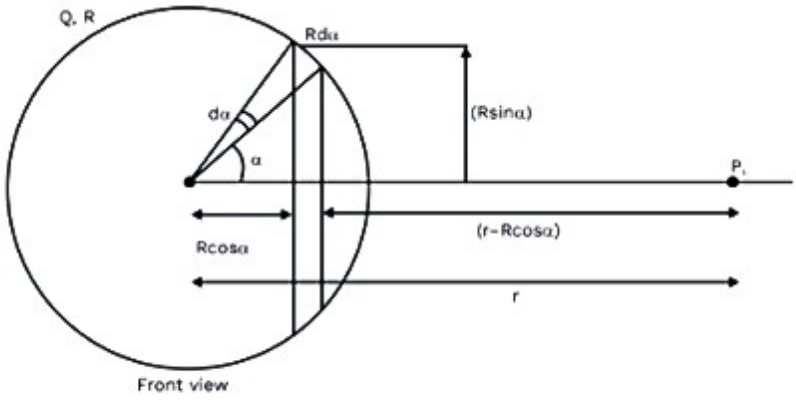
$$V \approx \frac{KQ}{r}$$

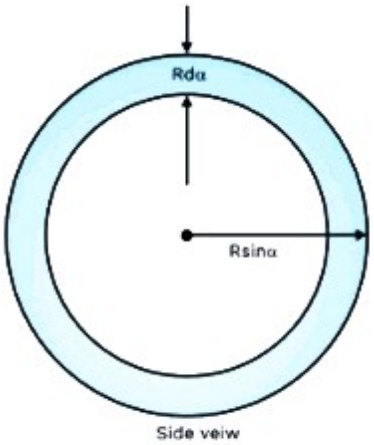
(Ring will behave like a point charge.)

The variation of electric potential can be plotted as:



Electric potential due to uniformly charged Spherical shell





Consider the shell to be made up by combination of infinitesimally thin rings as shown.

(a) Potential at a point outside the shell (at P_1)

$$\because \quad dV = \frac{K\,dq}{\sqrt{(r - R\cos\alpha)^2 + (R\sin\alpha)^2}}$$

$$\text{where } dq = \frac{Q}{4\pi R^2}\left(2\pi R\sin\alpha . R d\alpha\right) = \frac{Q}{2}\sin\alpha d\alpha$$

$$dV = \frac{KQ\sin\alpha\, d\alpha}{2\sqrt{r^2 + R^2 - 2rR\cos\alpha}}$$

Put $r^2 + R^2 - 2rR\cos\alpha = t^2$

Differentiating above equation w.r.t. α (here r and R are constants)-

$$0 - 2rR(-\sin\alpha) = \frac{2t\,dt}{d\alpha}$$

$$\sin\alpha\, d\alpha = \frac{t\,dt}{rR}$$

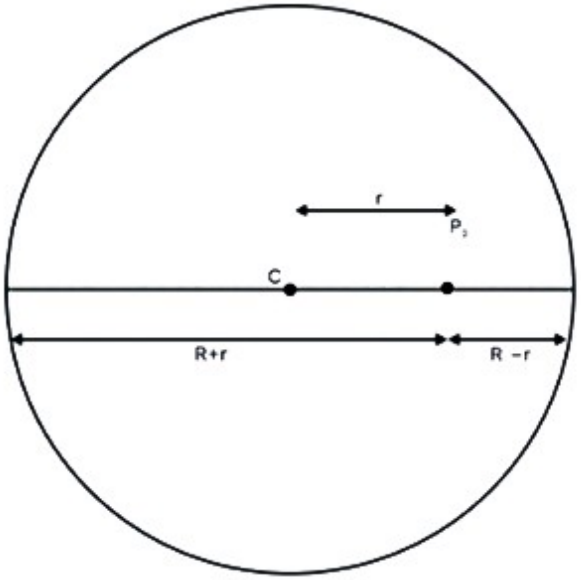
$$\therefore \quad dV = \frac{KQ}{2}\cdot\frac{t\,dt}{rRt} \Rightarrow dV = \frac{KQ}{2rR}dt$$

$$\Rightarrow \quad \int dV = \frac{KQ}{2rR}\int dt \Rightarrow V = \frac{KQ}{2rR}[t]$$

$$\Rightarrow \quad V = \frac{KQ}{2rR}\left[r^2 + R^2 - 2rR\cos\alpha\right]_{\alpha=0°}^{\alpha=180°}$$

$$\Rightarrow \quad V = \frac{KQ}{2rR}\left[(r+R) - (r-R)\right] \Rightarrow V = \frac{KQ}{2rR}(2R) \Rightarrow V_{out} = \frac{KQ}{r}$$

So, it can be said that for outside points, shell charge is acting like a point charge at the centre.



(b) Potential at a point inside the shell (at P_2)

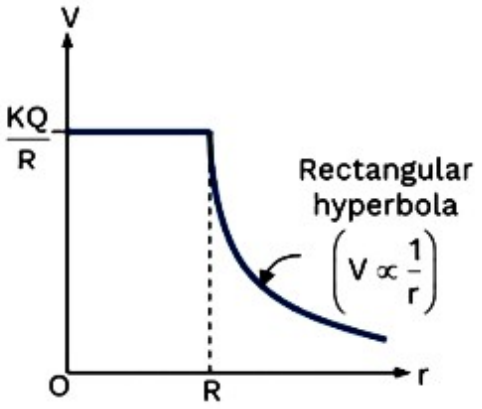
$$\because \quad V = \frac{KQ}{2rR}\left[r^2 + R^2 - 2rR\cos\alpha\right]_{\alpha=0^\circ}^{\alpha=180^\circ}$$

$$\therefore \quad V = \frac{KQ}{2rR}\left[(R+r) - (R-r)\right] = \frac{KQ}{2rR}(2r)$$

$$\Rightarrow \quad V_{in} = \frac{KQ}{R}$$

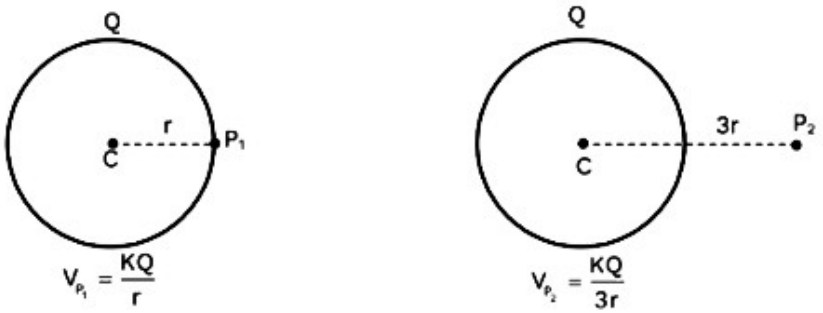
Inside the shell, potential is equal to the potential at the surface and is independent of r.

From the expression of V_{out} and V_{in}, we can plot the following graph.



Q. **A charged metal shell with radius r has the potential difference between its surface and a point at a distance 3r from the centre as V. What is the electric field at distance 3r from the centre?**

Sol:



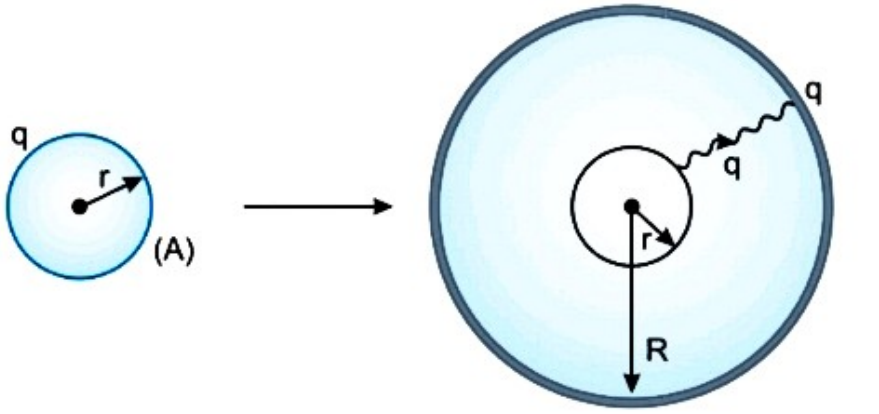
Given that, $V_{P_1} - V_{P_2} = V$

$$\therefore \frac{KQ}{r}\left(1-\frac{1}{3}\right) = V \Rightarrow \frac{2KQ}{3r} = V \Rightarrow Q = \frac{3Vr}{2K}$$

$$\text{Now, } E = \frac{KQ}{(3r)^2} = \frac{K}{9r^2}\left(\frac{3Vr}{2K}\right) = \frac{V}{6r}$$

Q. **Suppose there is a metal sphere (A) of radius 'r', charged to potential V. If the metal sphere is enclosed by a spherical shell B of radius R (R > r) and connected by a wire, then find the potential of the inner metal sphere.**

Sol:



If we connect inner sphere with outer, complete charge flows to outer surface of B

Before connecting the wire,

$$V = \frac{Kq}{r}$$

$$q = \frac{Vr}{K}$$

$$q = 4\pi \epsilon_0 \, rV$$

After connecting the conducting wire,

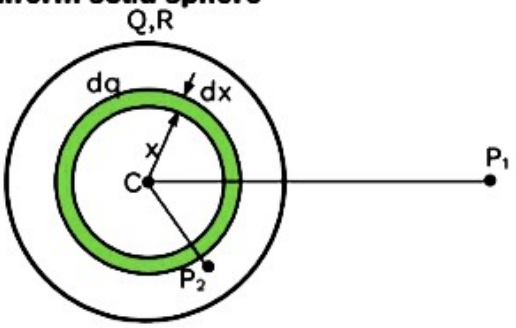
$$V_A = V_B = \frac{Kq}{R}$$

$$= \frac{K}{R} \times (4\pi \epsilon_0 \, rV)$$

$$= \frac{1}{4\pi \epsilon_0} \times \frac{1}{R} \times (4\pi \epsilon_0 \, rV)$$

$$\boxed{V_A = V_B = \frac{Vr}{R}}$$

Electric potential due to a uniform solid sphere



(a) Potential at a point outside the sphere ($OP_1 = r > R$)

Potential at P_1 due to shell of radius x, thickness dx is

$$dV = \frac{KdQ}{r} = \frac{K}{r}\left(\frac{Q}{\frac{4\pi}{3}R^3}.4\pi x^2 dx\right)$$

$$dV = \frac{3KQ}{rR^3}x^2 dx$$

$$\therefore \quad V = \frac{3KQ}{rR^3}\int_0^R x^2 dx = \frac{3KQ}{rR^3}.\frac{R^3}{3}$$

$$V_{out} = \frac{KQ}{r}$$

So, for the outside points, shell charge is acting like a point charge is kept at its centre.

(b) Potential at a point inside the sphere ($OP_2 = r < R$)

Potential at P_2 can be written as

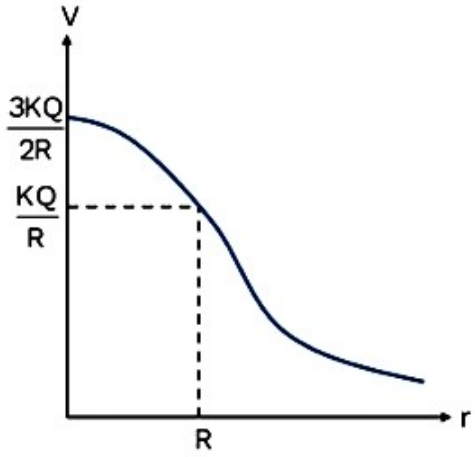
$$V = \int_0^r dV + \int_r^R dV = \int_0^r \frac{KdQ}{r} + \int_r^R \frac{KdQ}{x}$$

$$V = \int_0^r \frac{3KQ}{rR^3}.x^2dx + \int_r^R \frac{3KQ}{xR^3}.x^2dx$$

$$V = \left(\frac{3KQ}{rR^3}.\frac{r^3}{3}\right) + \left(\frac{3KQ}{R^3}.\frac{\left(R^2 - r^2\right)}{2}\right)$$

$$V = \frac{KQ}{R^3}\left(r^2 + \frac{3R^2}{2} - \frac{3r^2}{2}\right) = \frac{KQ}{R^3}\left(\frac{3R^2}{2} - \frac{r^2}{2}\right)$$

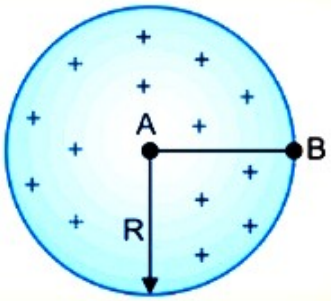
$$V_{in} = \frac{KQ}{2R^3}\left(3R^2 - r^2\right)$$



At centre (r = 0), $V_{centre} = \frac{3KQ}{2R} = \frac{3}{2}V_{surface}$

From the expression of V_{out} and V_{in}, we can plot the given graph.

Q. **Calculate the work done by electrostatic force in bringing a charge q from A to B in a non-conducting sphere having charge Q distributed through the volume of the sphere.**



Sol: From relation

$$W_{electric} = q(V_i - V_f)$$

$$W_{electric} = q(V_A - V_B)$$

We have seen that, for a non-conducting sphere,

$$V = \frac{KQ}{2R^3}\left(3R^2 - r^2\right)$$

At point 'A', r = 0

$$V_A = \frac{3}{2} \times \left(\frac{KQ}{R}\right)$$

At point 'B', r = R

$$V_B = \frac{KQ}{2R^3}\left(3R^2 - R^2\right) = \frac{KQ}{R}$$

So,

$$W_{electric} = q(V_A - V_B)$$

$$= q\left(\frac{3}{2}\frac{kQ}{R} - \frac{kQ}{R}\right)$$

$$= q \times \left(\frac{1}{2}\frac{kQ}{R}\right)$$

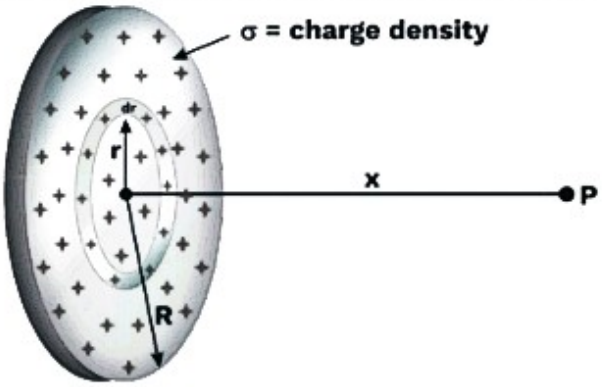
$$= \frac{1}{2} \times K \times \frac{q \times Q}{R}$$

$$= \frac{1}{2 \times 4\pi \epsilon_0} \times \frac{q \times Q}{R}$$

$$\boxed{W_{electric} = \frac{1}{8\pi \epsilon_0} \times \left(\frac{Qq}{R}\right)}$$

Electric potential on the axis of a charged disc

Consider a disc of uniform charge density σ as shown in the figure. To find the electric potential at a point P on the axis of the ring at a distance x from its centre, we consider the disc to be made up of a large number of infinitesimal rings of variable radii.



Electric potential due to elemental charged ring,

$$dV = \frac{K\,dq}{\sqrt{r^2 + x^2}} \qquad ...(1)$$

Charge $dq = \sigma(2\pi r dr)$

$$dV = \frac{K(\sigma)(2\pi r dr)}{\sqrt{r^2 + x^2}}$$

Since disc is combination of rings so total potential due to charged disc can be written as

$$V = \int dV = \int \frac{K(\sigma)(2\pi r dr)}{\sqrt{r^2 + x^2}}$$

$$V = \int dV = 2\pi\sigma K \int_0^R \frac{r\,dr}{\sqrt{r^2 + x^2}}$$

Substitute : $x^2 + r^2 = t^2$; $2r\,dr = 2t\,dt$

$$V = (2\pi K\sigma)\int \frac{t\,dt}{t}$$

$$= (2\pi\sigma K)[t]$$

$$= 2\pi\sigma K\left[\sqrt{x^2 + r^2}\right]_0^R$$

$$= 2\pi\sigma K\left[\sqrt{R^2 + x^2} - x\right]$$

$$= 2\pi\sigma \times \frac{1}{4\pi\,\epsilon_0}\left[\sqrt{R^2 + x^2} - x\right]$$

$$\boxed{V = \frac{\sigma}{2\,\epsilon_0}\left[\sqrt{R^2 + x^2} - x\right]}$$

σ : Surface charge density.

Q. **A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?**

Sol: If Q is the charge on sphere, then

$$V_{centre} = \frac{3KQ}{2R}$$

Half value of $V_{centre} = \frac{3KQ}{4R} = \frac{3}{4}V_{Surface} < V_{Surface}$

Therefore, we will use the expression of potential for outside the sphere.

$$V = \frac{KQ}{r} = \frac{3KQ}{4R} \Rightarrow r = \frac{4R}{3}$$

The distance from the surface = $\frac{4R}{3} - R = \frac{R}{3}$.

Change in Electrostatic Potential

If ΔU is the change in electrostatic potential energy between two points i and f, then the corresponding change in electric potential ΔV between same points is given by

$$\Delta V = \lim_{q_0 \to 0}\left(\frac{\Delta U}{q_0}\right) \quad \text{or} \quad V_f - V_i = \lim_{q_0 \to 0}\left(\frac{U_f - U_i}{q_0}\right)$$

Q. **A charge of 5C is given a displacement of 0.5 m and the work done in the process is 10J. The potential difference between the two points will be**

Sol: Potential difference, $\Delta V = \frac{\Delta U}{q_0} = \frac{W_{ext}}{q_0} = \frac{10}{5} = 2\,\text{volt}$

Q. **If a charge is shifted from a low-potential region to a high-potential region, the electric potential energy**

(A) increases **(B) decreases**

(C) remains the same **(D) may increases or decreases**

Sol: $\because \quad \Delta U = q_0\,\Delta V$

Given $\Delta V = V_{high} - V_{low} > 0$

Since q_0 can be +ve or −ve, therefore ΔU can be +ve or −ve. In other words, ΔU may increase or decrease.

Q. **20J of work has to be done against an existing electric field to take a charge of −0.1 C from A to B, then potential difference $V_B - V_A$ is**

Sol: Potential difference, $V_B - V_A = \left(\frac{\Delta U}{q_0}\right) = \left(\frac{W_{A\to B}}{q_0}\right) = \frac{20}{-0.1} = -200\text{V}$

Q. **Two points P and Q are maintained at the potentials of 10V and −4V respectively. The work done in moving 100 electrons from P and Q is**

Sol: Change in potential energy of the electrons,

$$\Delta U = Q_{ext} = q_0\left(V_Q - V_P\right) = 100\left(-1.6 \times 10^{-19}\right)\left[\left(-4\right) - \left(10\right)\right]$$

$W_{P\to Q} = 2.24\times10^{-16}\,J$

Work done here will be positive as the external agent is moving negative charges from higher to lower potential.

Q. 12J of work has to be done against an existing electric field to take a charge of 0.01 C from A to B. How much is the potential difference $V_B - V_A$?

Sol: Potential difference, $(V_B - V_A) = \frac{W_{A\to B}}{q_0} = \frac{12}{0.01} = 1200V$

Relation between electric field (E) and potential (V)

(a) Potential in terms of electric field

A small change in electrostatic potential V is defined as

$$dV = \frac{dU}{q} = \frac{-dW_{elec}}{q} = -\frac{\vec{F}_{elec}}{q}.d\vec{r}$$

$$dV = -\vec{E}.d\vec{r}$$

∴ Change in potential between initial point (i) to final point (f) is

$$\int_i^f dV = -\int_i^f \vec{E}.d\vec{r}$$

$$\boxed{V_f - V_i = -\int_i^f \vec{E}.d\vec{r}}$$

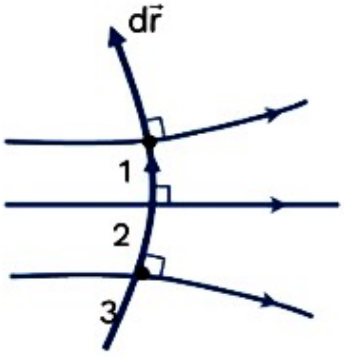
Note:
Electric potential remains constant when we move on a plane perpendicular to electric field lines.

In other words,

if $\vec{E} \perp d\vec{r}$,

then $dV = -\vec{E}.d\vec{r} = 0$

or $V_1 = V_2 = V_3$.

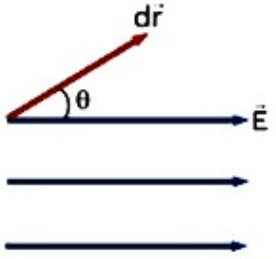


(b) Electric field in terms of potential

(i) While moving at an angle θ to the $\vec{E}$,

$$dV = -\vec{E}.d\vec{r} = -Edr\cos\theta$$

$$\frac{dV}{dr} = -E\cos\theta = -E_r$$



(Rate of change of V w.r.t. r along the direction of $\vec{r}$) = – (Component of $\vec{E}$ along the direction of $\vec{r}$).

Here, we can say, along the direction of $\vec{E}\left(\theta = 0^{\circ}\right)$, potential decreases with maximum rate and is equal to E.

Similarly, $\left|\frac{dV}{dr}\right|$ will be minimum (zero) when $\theta = 90^{\circ}$.

(ii) If $\vec{E} = E_x\hat{i} + E_y\hat{j} + E_z\hat{k}$ and

$d\vec{r} = dx\hat{i} + dy\hat{j} + dz\hat{k}$, then

$$dV = -\vec{E}.d\vec{r} = -\left(E_x dx + E_y dy + E_z dz\right)$$

If particle moves along x-axis, (dy = dz = 0 and $E_x = -\frac{dV}{dx}$)

or $E_x = -\left(\frac{\partial V}{\partial x}\right)$ = – (differentiation of V w.r.t x by keeping y and z constant).

E_x = – (Partial differentiation on of V w.r.t x only)
Similarly,

$E_y = -\left(\frac{\partial V}{\partial y}\right)$ = – (Partial differentiation of V w.r.t. y only)

and $E_z = -\left(\frac{\partial V}{\partial z}\right)$ = – (Partial differentiation of V w.r.t. z only)

So, we can write

$$\vec{E} = -\left(\frac{\partial V}{\partial x}\hat{i} + \frac{\partial V}{\partial y}\hat{j} + \frac{\partial V}{\partial z}\hat{k}\right)$$

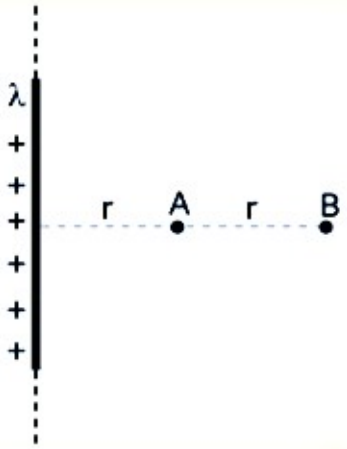
$$\vec{E} = -\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)V = -\text{grad}\left(V\right) = \nabla\left(V\right)$$

Q. **A hollow spherical conductor of radius R is given a charge Q. Work done in moving a charge q from its centre to surface is**

Sol: We know that the electric field inside a charged conducting shell = 0

$$\therefore\ \Delta W = \Delta U = -\int_{\text{centre}}^{\text{surface}} q\vec{E}.d\vec{r} = 0$$

Q. **Find $V_B - V_A$ for the given infinitely long line charge of linear charge density λ.**



Sol: Using the relation, $V_B - V_A = -\int_A^B \vec{E}.d\vec{r}$

$$V_B - V_A = -\int_A^B \vec{E}.d\vec{r} = -\frac{\lambda}{2\pi\varepsilon_0}\int_r^{2r}\frac{dr}{r} = \frac{-\lambda}{2\pi\varepsilon_0}\left[\ln r\right]_r^{2r} = -\left(\frac{\lambda \ln 2}{2\pi\varepsilon_0}\right)$$

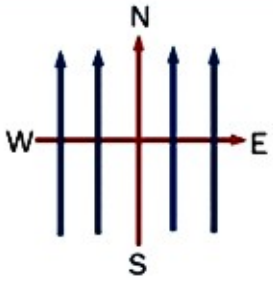
Q. **At a point in space, the electric field points towards north. In the region surrounding this point, the rate of change of potential will be zero along**
(A) North **(B) South** **(C) North-South** **(D) East-West**

Sol: **(D)**

Rate of change of potential, $\frac{dV}{dr} = 0$.

or V = constant

V can be constant in plane perpendicular to $\vec{E}$,
out of given options, East-West can be such direction.

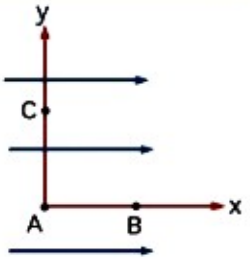


Q. **A uniform electric field pointing in positive x-direction exists in a region. Let A be the origin, B be the point on the x-axis at x = + 1 cm and C be the point on the y-axis at y = + 1 cm. Then the potential at the points A, B and C satisfy**
(A) $V_A > V_B$ **(B) $V_A < V_C$** **(C) $V_A < V_B$** **(D) $V_A > V_C$**

Sol: **(A)**

$\because$ Along the direction of electric field, the electric potential decreases.

$\therefore V_A = V_C > V_B$



Q. **When a charge of 3C is placed in a uniform electric field, it experiences of a force of 3000 N. The potential difference between two points separated by a distance of 1 cm along the field is**

Sol: $\because \quad E = \frac{F}{q} = \frac{3000}{3} = 1000\,Vm^{-1}$

Along the field, $\theta = 0°$

Also, $\Delta r = 1cm = 10^{-2}\,m$

$\therefore \quad |\Delta V| = E\,\Delta r \cos\theta = 1000\left(10^{-2}\right)\cos 0°$

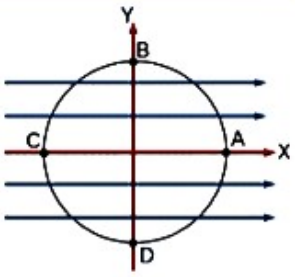
$|\Delta V| = 10V$

Q. **The electric field at the origin is along the positive x-axis. A small circle is drawn with the centre at the origin cutting the axes at points A, B, C and D having coordinates (a, 0), (0, a), (–a, 0), (0, –a) respectively. Out of the points on the periphery of the circle, the potential is minimum at**

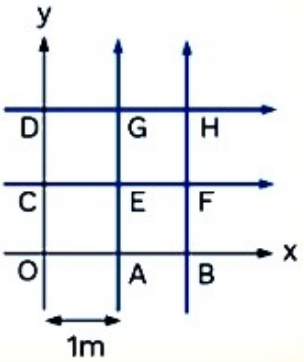
(A) A **(B) B** **(C) C** **(D) D**

Sol: **(A)**

As we can see in diagram, point A is ahead (as compared to the other point) along the field. Therefore, it will have the minimum potential.



Q. **The grid (each square of 1 m × 1 m) represents a region in space containing a uniform electric field. If potentials at point O, A, B, C, D, E, F, G, H are respectively 0, –1, –2, 1, 2, 0, –1, 1 and 0 volts, then find the electric field intensity.**



(A) $\left(\hat{i} + \hat{j}\right) V/m$ **(B)** $\left(\hat{i} - \hat{j}\right) V/m$

(C) $\left(-\hat{i} + \hat{j}\right) V/m$ **(D)** $\left(-\hat{i} - \hat{j}\right) V/m$

Sol: **(B)**

Along the x-axis,

$$E_x = \frac{-\Delta V}{\Delta x} = \frac{-(V_B - V_A)}{(x_B - x_A)} = \frac{-[(-2)-(-1)]}{1} = 1\ V/m$$

Along the y-axis,

$$E_y = \frac{-\Delta V}{\Delta y} = \frac{-(V_D - V_C)}{(y_D - y_C)} = \frac{-[-2-1]}{1} = -1\ V/m$$

$\therefore$ Electric field vector, $\vec{E} = E_x\hat{i} + E_y\hat{j}$ or $\vec{E} = (\hat{i} - \hat{j})\ V/m$

Q. **Let V_0 be the potential at the origin in an electric field $\vec{E} = E_x\hat{i} + E_y\hat{j}$. The potential at the point (x, y) is**

Sol: $\because \int_{(0,0)}^{(x,y)} dV = -\int \vec{E}.d\vec{r} = -\int_{(0,0)}^{(x,y)} (E_x\hat{i} + E_y\hat{j}).(dx\hat{i} + dy\hat{j})$

$V_{(x,y)} - V_{(0,0)} = -[E_x x + E_y y]_{(0,0)}^{(x,y)}$

$V_{(x,y)} - V_0 = -E_x x - E_y y$

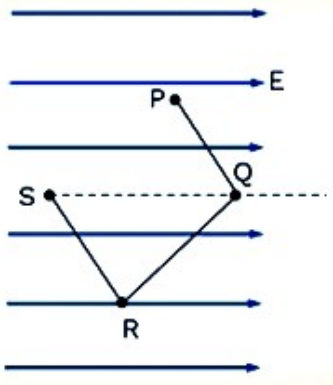
$V_{(x,y)} = V_0 - E_x x - E_y y$

Q. **Charge Q is given a displacement $\vec{r} = a\hat{i} + b\hat{j}$ in an electric field $\vec{E} = E_1\hat{i} + E_2\hat{j}$. The work done by electric force is**

Sol: $W_{electric} = \vec{F}_{electric}.\vec{r} = Q\vec{E}.\vec{r} = Q(E_1 i + E_2\hat{j}).(a\hat{i} + b\hat{j})$

$W_{electric} = Q(E_1 a + E_2 b)$

Q. **Point charge (q) moves from point (P) to point (S) along the path PQRS as shown in figure in a uniform electric field E. Pointing co-parallel to the positive direction of the x-axis. The co-ordinates of the points P, Q, R and S are (a, b, 0), (2a, 0, 0), (a, –b, 0) and (0, 0, 0) respectively. The work done by the field in the above process is given by the expression**



(A) q E b **(B) –q E a**

(C) q E a **(D)** $qE\sqrt{[(2a)^2 + (b)^2]}$

Sol: **(B)**

$$W_{electric} = q\left(\vec{E}.\Delta\vec{r}\right) = q\left(E\hat{i}\right).\left(\vec{r}_s - \vec{r}_p\right) = q\left(E\hat{i}\right).\left[\left(0\hat{i} + 0\hat{j} + 0\hat{k}\right) - \left(a\hat{i} + b\hat{j} + 0\hat{k}\right)\right]$$

$$W_{electric} = q\left(E\hat{i}\right).\left(-a\hat{i} - b\hat{j}\right)$$

$$\Rightarrow W_{electric} = -qEa$$

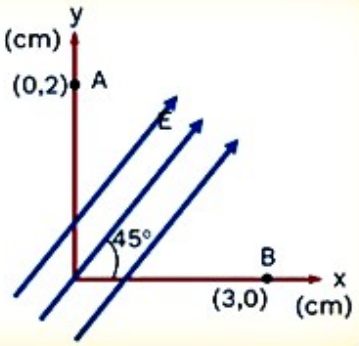
Q. **A uniform electric field of 400 V/m is directed at 45° above the x-axis as shown in figure. The potential difference V_A-V_B is given by**

(A) 0 V

(B) 4 V

(C) 6.4 V

(D) 2.8 V



Sol: **(D)**

$$V_A - V_B = -\int_B^A \vec{E}.d\vec{r} = -\int_{(0.03,0)}^{(0,0.02)} 400\left(\frac{1}{\sqrt{2}}\hat{i} + \frac{1}{\sqrt{2}}j\right).\left(dx\hat{i} + dy\hat{j}\right)$$

$$V_A - V_B = -200\sqrt{2}\left[x + y\right]_{(0.03,0)}^{(0,0.02)}$$

$$V_A - V_B = -200\sqrt{2}\left[\left(0 + 0.02\right) - \left(0.03 + 0\right)\right]$$

$$V_A - V_B = 2\sqrt{2} = 2.8\,V$$

Q. **The electric potential V at any point x, y, z in space is given by $V = 4x^2$ volt. The electric field at the point (1 m, 0, 2 m) is**

Sol: Electric field, $\vec{E} = -\text{grad}.V = -\left[\frac{\partial\left(4x^2\right)}{\partial x}\hat{i} + \frac{\partial}{\partial y}\left(4x^2\right)\hat{j} + \frac{\partial\left(4x^2\right)}{\partial z}\hat{k}\right]$

$$\vec{E} = -8x\hat{i} = -8\left(1\right)\hat{i} = -8\hat{i}$$

$$\left|\vec{E}\right| = 8\,V\,m^{-1}$$

Q. **At any point (x, 0, 0), the electric potential V is $\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}\right)$ volt, then electric field intensity at x = 1 m is**

Sol: $E_x = -\frac{\partial V}{\partial x} = \frac{-\partial}{\partial x}\left(\frac{1000}{x} + \frac{1500}{x^2} + \frac{500}{x^3}\right)$

$E_x = -\left[\frac{-1000}{x^2} - \frac{3000}{x^3} - \frac{1500}{x^4}\right] = \left[1000 + 3000 + 1500\right] = 5500\,V\,m^{-1}$

Similarly, $E_y = 0$ and $E_z = 0$

$\therefore \vec{E} = 5500\,\hat{i}\ V/m$

Q. **Electric potential at a point is $V = x^2y + yz$. The electric field at the point (1, 3, 1) is**

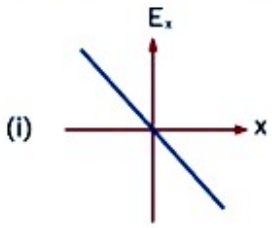
Sol: Electric field, $\vec{E} = -\left[\frac{\partial}{\partial x}\left(x^2y + yz\right)\hat{i} + \frac{\partial}{\partial y}\left(x^2y + yz\right)\hat{j} + \frac{\partial}{\partial z}\left(x^2y + yz\right)\hat{k}\right]$

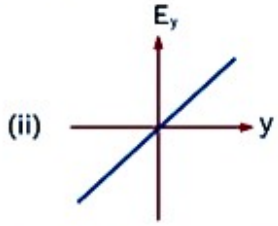
So, $\vec{E} = -\left[2xy\hat{i} + \left(x^2 + z\right)\hat{j} + y\hat{k}\right]$

At point (1, 3, 1),

$\vec{E} = -\left(6\hat{i} + 2\hat{j} + 3\hat{k}\right)$

Q. **The potential of a field is given by equation $V = x^2 - y^2$. The best representation of electric field with x and y axis is**





(A) only (i) is correct
(B) only (ii) is correct
(C) both (i) and (ii) are incorrect
(D) both (i) & (ii) are correct

Sol: **(D)**

$\because E_x = \frac{-\partial V}{\partial x} = \frac{-\partial}{\partial x}\left(x^2 - y^2\right) = -2x$

$\therefore E_x - x$ graph will be straight line with negative slope so, graph (i) is correct.

Similarly, $E_y = \frac{-\partial}{\partial y}\left(x^2 - y^2\right) = 2y$

So, graph (ii) is also correct.

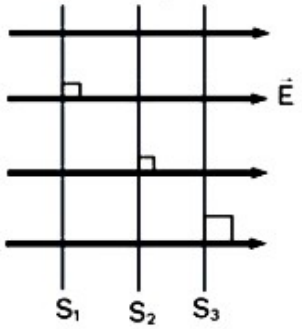
Q. **A non-conducting ring of radius 0.5 m carries a total charge of 1.11×10^{-10}C distributed non uniformly on its circumference producing an electric field E everywhere in space. The value of the integral $\int_{\ell=\infty}^{\ell=0} -E.d\ell$ (ℓ = 0 being centre of the ring) in volts is**

Sol: $$\int_{\ell=\infty}^{\ell=0} -\vec{E}.d\vec{\ell} = \int_{\infty}^{0} dV = [V]_{\infty}^{0} = \left[\frac{Kq}{\sqrt{R^2 + r^2}}\right]_{\infty}^{0}$$

So, $$\int_{\ell=\infty}^{\ell=0} -\vec{E}.d\vec{\ell} = \frac{Kq}{R} - 0 = \frac{9 \times 10^9 \times 1.11 \times 10^{-10}}{0.5} = 1.998\,V$$

Equipotential Surface

A surface at every point of which electric potential is same is called an equipotential surface. In other words, it is the locus of all points on which potential is constant.



Surfaces S_1, S_2 and S_3 shown in the above figure are equipotential surfaces.

Also, $V_1 > V_2 > V_3$.

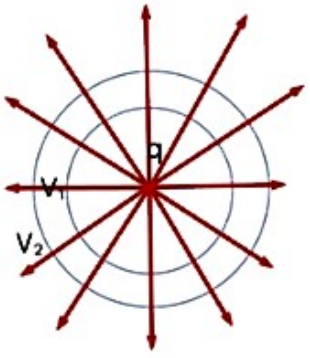
Important properties of equipotential surfaces

(1) For uniform electric field, the equipotential surfaces are a family of parallel planes perpendicular to field lines.

(2) The equipotential surfaces produced by a point charge are a family of concentric spheres.

If $q > 0$, $V_1 > V_2$

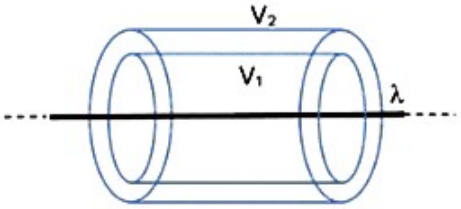
If $q < 0$, $V_1 < V_2$



(3) Equipotential surfaces are co-axial cylinders for a linear charge distribution.

If $\lambda > 0$, $V_1 > V_2$

If $\lambda < 0$, $V_1 < V_2$

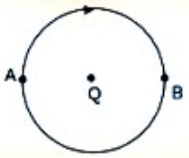


(4) Two equipotential surfaces cannot intersect each other. If they do so, then the intersection line will have two potentials which is not possible.

(5) Direction of electric field is perpendicular to the equipotential surfaces.

(6) Perpendicular to the electric field, electrostatic potential V remains constant.

Q. **A charge Q is placed at the centre of a circle of radius R. The work done in moving a charge q from A to B so as to complete a semicircle is**



Sol: ∵ A and B are present on an equipotential surface.

$\therefore V_B - V_A = 0 \Rightarrow W_{A \to B} = q(V_B - V_A) = 0$

$\Rightarrow W_{A \to B} = 0$

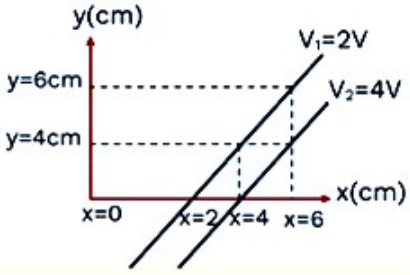
Q. **Figure shows two equipotential surfaces V_1 & V_2. The component of electric field in x & y direction will be**

(A) $E_x\hat{i}$ **and** $E_y\hat{j}$

(B) $-E_x\hat{i}$ **and** $E_y\hat{j}$

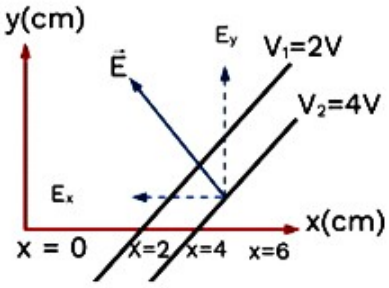
(C) $E_x\hat{i}$ **and** $-E_y\hat{j}$

(D) $-E_x\hat{i}$ **and** $-E_y\hat{j}$



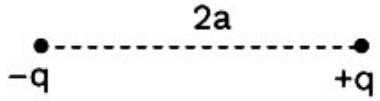
Sol: **(B)**

Since, field lines are perpendicular to the equipotential surfaces and also flow from high potential to low potential. Therefore, we can draw the electric field as shown. This shows the x component of electric field is along negative x axis and y component is along positive y axis.



Electric Dipole

Electric dipole is an arrangement of two equal and opposite charges separated by a distance.



Consider that two point charges +q and -q are separated by a distance 2a as shown in the figure. It is an electric dipole.

Dipole Moment $(\vec{p})$

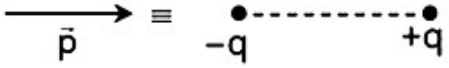
Dipole moment is the product of magnitude of either charge (q) and distance between the charges (2a).

$$\vec{p} = q(2\vec{a})$$

(1) Dipole moment is a vector quantity, and it is directed from negative to positive charge.

(2) The dimension of p is, $[p] = [L^1T^1A^1]$.

(3) Symbol of an electric dipole is as shown.

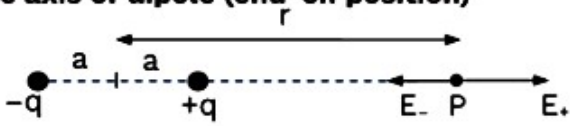


(4) Examples: Polar molecules like NaCl, HCl, H_2O, etc.

(5) Polar molecules have some dipole moment because the centre of positive and negative charges does not coincide while non-polar molecules do not have any dipole moment as the centre of positive and negative charge coincides at the same point. CO_2, CH_4, etc are non-polar molecules.

Electric Field due to an electric dipole

(A) Electric field on the axis of dipole (end-on position)



For a point P lying on the axis of the dipole as shown,

$$E_{axis} = E_+ - E_- = \frac{Kq}{(r-a)^2} - \frac{Kq}{(r+a)^2}$$

$$E_{axis} = Kq\frac{4ra}{(r^2-a^2)^2} = \frac{2K[q(2a)]}{(r^2-a^2)^2}$$

$$E_{axis} = \frac{2Kp}{(r^2-a^2)^2}$$

(along the axis of dipole)

If $r >> a$, $\vec{E}_{axis} \approx \frac{2K\vec{p}}{r^3}$

(B) Electric field on the equatorial line (broadside-on position)

$$E_{eq} = 2E_+ \cos\theta$$

$$E_{eq} = \frac{2Kq}{(r^2+a^2)} \cdot \frac{a}{\sqrt{r^2+a^2}}$$

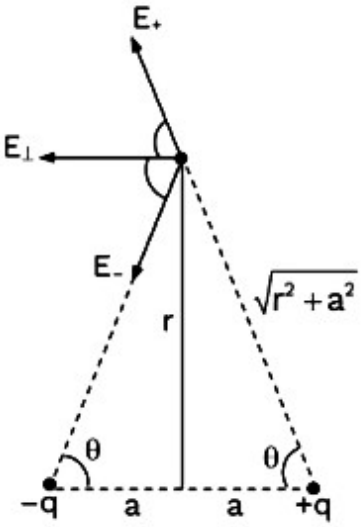
$$E_{eq} = \frac{Kp}{(r^2+a^2)^{3/2}}$$

(opposite to direction of dipole moment).

$$\vec{E}_{eq} = \frac{-K\vec{p}}{(r^2+a^2)^{3/2}}$$

If $r >> a$,

$$\vec{E}_{eq} = \frac{-K\vec{p}}{r^3}$$



The ratio of the electric field due to an electric dipole on its axis and on the perpendicular bisector of the dipole (at distant point) is

$$\left|\frac{E_{axis}}{E_\perp}\right| = \frac{2Kp/r^3}{Kp/r^3} = \frac{2}{1}$$

(C) Electric field at a general point (r, θ)

$$E_\parallel = \frac{2K(p\cos\theta)}{r^3} \quad ...(i)$$

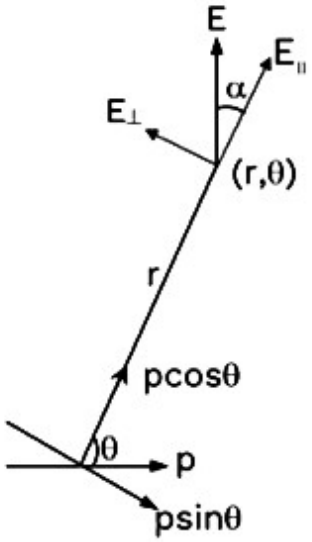
and $E_\perp = \frac{K(p\sin\theta)}{r^3}$ ...(ii)

Resultant of $E_\parallel$ and $E_\perp$,

$$E = \sqrt{E_\parallel^2 + E_\perp^2}$$

$$E = \frac{Kp}{r^3}\sqrt{4\cos^2\theta + \sin^2\theta}$$

$$E = \frac{Kp}{r^3}\sqrt{3\cos^2\theta + 1}$$



Direction of E or the angle between resultant field E and radial axis :

$$\tan\alpha = \frac{E_\perp}{E_\parallel} = \frac{\tan\theta}{2} \Rightarrow \alpha = \tan^{-1}\left(\frac{\tan\theta}{2}\right)$$

Here direction of E from the dipole axis is $(\alpha + \theta)$.

Q. **A short dipole is placed along x-axis with centre at origin. The electric field at a point P, which is at a distance r from origin such that $\overline{OP} = \vec{r}$ makes an angle of 45° with x-axis, is directed along a direction making-**

(A) $\tan^{-1}(0.5)$ **with x-axis**

(B) $\pi/4 + \tan^{-1}(0.5)$ **with x-axis**

(C) $\pi/4 + \tan^{-1}(0.5)$ **with y-axis**

(D) $\tan^{-1}(0.5)$ **with y-axis**

Sol: **(B)**

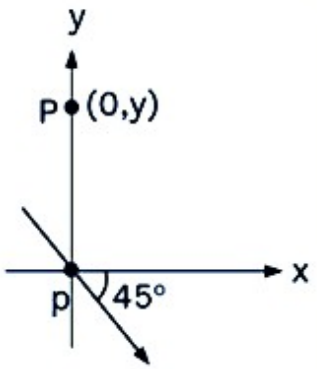
The angle α with the radial axis

$$\tan\alpha = \frac{\tan\theta}{2} = \frac{\tan 45^\circ}{2} = 0.5$$

$$\alpha = \tan^{-1}(0.5)$$

The angle with the x-axis = $\frac{\pi}{4} + \tan^{-1}(0.5)$.

Q. **A dipole is placed at origin of coordinate system as shown in figure, find the electric field at point P(0, y).**



(A) $\frac{Kp}{y^3}\left(-\hat{i} - 2\hat{j}\right)$

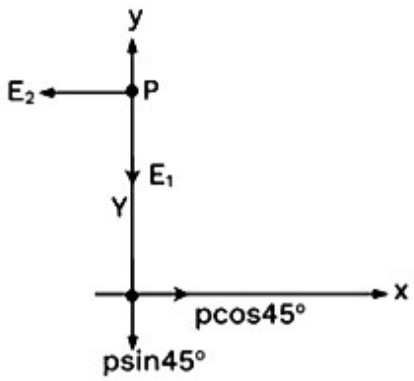
(B) $\frac{Kp}{\sqrt{2}y^3}\left(\hat{i} + 2\hat{j}\right)$

(C) $\frac{Kp}{\sqrt{2}y^3}\left(-\hat{i} - 2\hat{j}\right)$

(D) none of these

Sol: **(C)**

We can resolve the dipole moment vector along x and y axes as shown.



Electric field at P due to psin45° is E_1 given as,

$$E_1 = \frac{2K\left(p\sin 45^\circ\right)}{y^3} = \frac{2Kp}{\sqrt{2}y^3}$$

Electric field due to pcos45° is E_2 given as,

$$E_2 = \frac{K\left(p\cos 45^\circ\right)}{y^3} = \frac{Kp}{\sqrt{2}y^3}$$

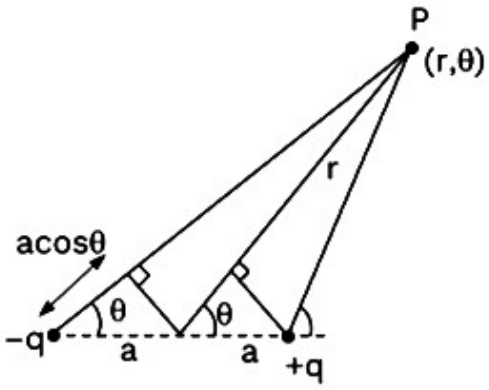
Net electric field at point P,

$$\vec{E}_P = -E_2\hat{i} - E_1\hat{j}$$

$$\vec{E}_p = \frac{Kp}{\sqrt{2}y^3}\left(-\hat{i} - 2\hat{j}\right)$$

Electric Potential due to an electric dipole

For r > > a, all three lines meeting at P will be approximately parallel.
In this case,
Distance between −q and P = r + acosθ
Distance between +q and P = r − acosθ
Potential at P = V = $V_+ + V_-$



$$V = \frac{Kq}{r - a\cos\theta} - \frac{Kq}{r + a\cos\theta} = \frac{Kq(2a\cos\theta)}{r^2 - a^2\cos^2\theta}$$

$$V = \frac{Kp\cos\theta}{r^2 - a^2\cos^2\theta}$$

$$V \approx \frac{Kp\cos\theta}{r^2} \text{ or } V = \frac{K\vec{p}.\vec{r}}{r^3}$$

For a point on equatorial plane (θ = 90°), $V_{90°} = 0$.

For a point on equatorial plane (θ = 0°), $V_{0°} = \frac{Kp}{r^2}$ = maximum.

For a point on equatorial plane (θ = 180°), $V_{180°} = \frac{-Kp}{r^2}$ = minimum.

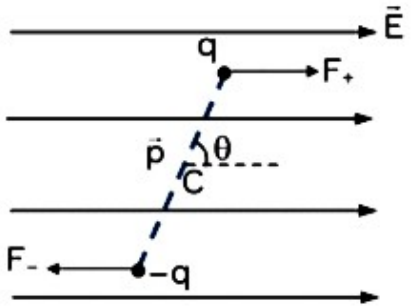
Torque on a dipole in a uniform electric field

On placing an electric dipole in uniform electric field at an angle θ, the forces acting on the charges are,

$F_+ = F_- = qE$

$\therefore \quad \vec{F}_{dipole} = \vec{F}_+ + \vec{F}_- = 0$

Torque acting on the dipole, $\vec{\tau} = \vec{r} \times \vec{F}$



$$\vec{\tau}_{dipole} = (2a)qE\sin\theta \otimes$$

$$\vec{\tau}_{dipole} = pE\sin\theta \otimes$$

$$\Rightarrow \vec{\tau}_{dipole} = \vec{p} \times \vec{E}$$

Q. An electric dipole is placed at an angle of 30° with an electric field of intensity 2×10^5 N/C. It experiences a torque equal to 4 N-m. Calculate the charge on the dipole if the dipole length is 2 cm.

Sol: Torque, $\tau = pE\sin\theta = (2a)qE\sin\theta$

$$\therefore \quad q = \frac{\tau}{(2a)E\sin\theta} = \frac{4}{\left(2\times10^{-2}\right)\times2\times10^{5}\times\sin30^{\circ}} = 2\times10^{-3}$$

Magnitude of each charge on the dipole, q = 2 mC.

Potential Energy of electric dipole in electric field

External work done on the dipole in rotating it by an angle $d\theta$,

$$dW_{ext} = \vec{\tau}_{elec}.d\vec{\theta}$$

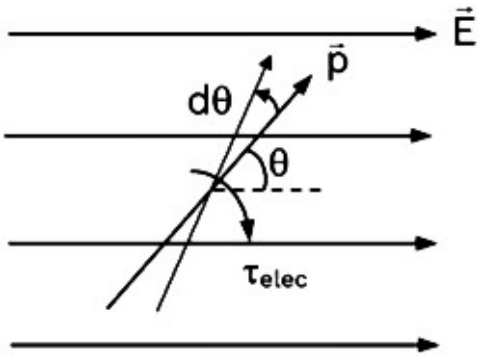
$$\therefore \quad dU = -\vec{\tau}_{elec}.d\vec{\theta}$$

$$\therefore \quad \int_{i}^{f} dU = -\int_{\theta_i}^{\theta_f} \vec{\tau}_{elec}.d\vec{\theta}$$

$$\Rightarrow \quad U_f - U_i = -\int_{\theta_i}^{\theta_f} pE\sin\theta d\theta\cos180^{\circ}$$

$$\Rightarrow \quad U_f - U_i = pE\left[-\cos\theta\right]_{\theta_i}^{\theta_f}$$

$$\Rightarrow \quad U_f - U_i = -pE\left(\cos\theta_f - \cos\theta_i\right)$$



If we take initial point (i) as a reference point at $\theta = 90^{\circ}$ and assign $U_{90^{\circ}} = 0$, also take final point (f) at general angle (θ), then

$$U_\theta - U_{90^{\circ}} = -pE\left[\cos\theta - \cos90^{\circ}\right]$$

$$U_\theta = -pE\cos\theta$$

$$U_\theta = -\vec{p}.\vec{E}$$

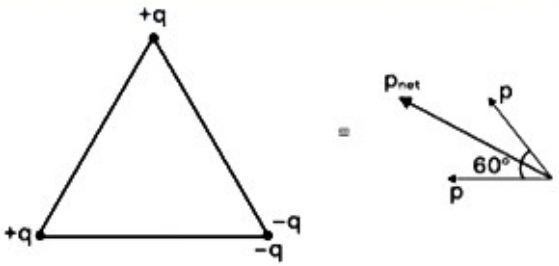
If $\theta = 0^{\circ} \Rightarrow U_\theta = -pE$ (minimum P.E.)
If $\theta = 180^{\circ} \Rightarrow U_\theta = +pE$ (maximum P.E.)

Q. Electric charges q, q and –2q are placed at the corners of an equilateral triangle ABC of side ℓ. The magnitude of electric dipole moment of the system is

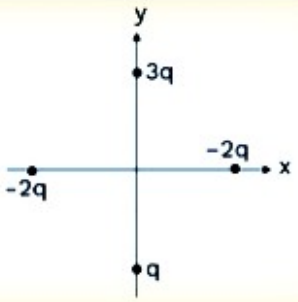
Sol: -2q charge can be taken as -q and -q. So, we get two electric dipoles as shown.

$$p_{net} = \sqrt{p^2 + p^2 + 2p^2\cos60^{\circ}} = \sqrt{3}\,p$$

$$p_{net} = \sqrt{3}\,q\ell$$



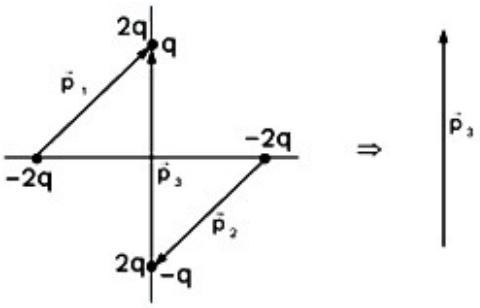
Q. **Four charges are placed at a distance 'a' from origin. The dipole moment of configuration is**



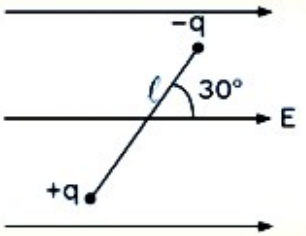
Sol: The breakup of the given charges is as shown,

$$\vec{p}_{net} = (\vec{p}_1 + \vec{p}_2) + \vec{p}_3 = \vec{0} + \vec{p}_3 = q(2a)\,\hat{j}$$

$$\vec{p}_{net} = (2qa)\,\hat{j}$$



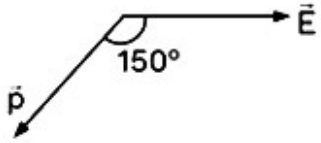
Q. **The torque and potential energy of the given configuration of dipole is**



Sol: Torque on the dipole, $\vec{\tau} = \vec{p} \times \vec{E}$

$\therefore \quad \vec{\tau} = q\ell E \sin 150^\circ \odot$

$$\vec{\tau} = \frac{q\ell E}{2} \odot$$



Also, potential energy, $U = -\vec{p}.\vec{E}$

$$U = -q\ell E \cos 150^\circ = \frac{\sqrt{3}\,q\ell E}{2}$$

Q. **Draw the graph of potential energy (U) versus θ (angle between $\vec{p}$ and $\vec{E}$) for an electric dipole present in a uniform electric field. Also comment on the stability of dipole.**

Sol: Potential energy of dipole, $U = -(\vec{p}.\vec{E})$

$\therefore \quad U = -pE\cos\theta$

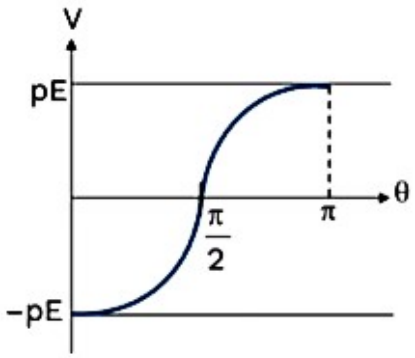
$U \propto (-\cos\theta)$

When $\theta = 0°$, $\tau = 0$ and $U_{min} = -pE$

$\Rightarrow$ Dipole will be in stable equilibrium

When $\theta = 180°$, $\tau = 0$ and $U_{max} = +pE$

$\Rightarrow$ Dipole will be in unstable equilibrium



Q. **Find the work done by external force in deflecting a dipole slowly from stable equilibrium position to unstable equilibrium position. Dipole has moment (p) and it is present in uniform electric field E.**

Sol: $W_{ext} = \Delta U = U_{180°} - U_{0°} = (-pE\cos 180°) - (-pE\cos 0°)$

$\Rightarrow W_{ext} = 2pE$

Q. **An electric dipole is placed parallel to the electric field. If W is the work done in rotating the dipole by 60° (from stable equilibrium position), then work done in rotating it by 180° (from stable equilibrium position).**

Sol: $\because \quad W = \Delta U = U_{60°} - U_{0°} = (-pE\cos 60°) - (-pE\cos 0°)$

$$W = \frac{-pE}{2} + pE = \frac{pE}{2}$$

$\therefore W_{0° \to 180°} = W' = U_{180°} - U_{0°} = (-pE\cos 180°) - (-pE\cos 0°)$

$\Rightarrow \quad W' = 2pE \quad \Rightarrow W' = 4W.$

Q. **An electric dipole is placed at an angle of 30° to a non-uniform electric field. The dipole will experience**

(A) a torque as well as a translational force

(B) a torque only

(C) a translational force only in the direction of the field

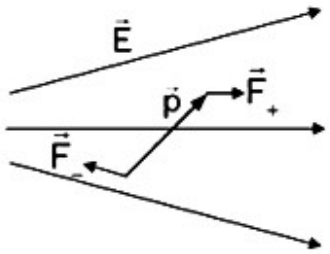
(D) a translational force only in a direction normal to the direction of the field

Sol: **(A)**

A non-uniform is shown here,
we can clearly observe

$$\left(\sum \vec{F}\right)_{dipole} = \vec{F}_+ + \vec{F}_- \neq \vec{0}$$

Also, $\left(\sum \vec{\tau}\right)_{dipole} \neq \vec{0}$



Force between two short dipoles

(a) When dipoles are placed on the same axis

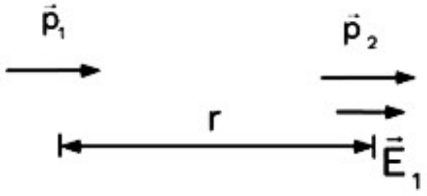
Assume r >> dipole size.

Since, field $\vec{E}_1$ due to $\vec{p}_1$ around $\vec{p}_2$ is non-uniform, therefore $\vec{p}_2$ will experience force.

The interaction energy of $\vec{p}_2$ and $\vec{E}_1$ is,

$$U = -\vec{p}_2 . \vec{E}_1 = -p_2 \frac{2Kp_1}{r^3}\cos 0°$$

$$U = -\frac{2Kp_1p_2}{r^3}$$



Force on dipole $\vec{p}_2$ is, $\vec{F} = -\frac{dU}{dr}\hat{r}$

$$\vec{F} = +2Kp_1p_2 \frac{d}{dr}\left(\frac{1}{r^3}\right)\hat{r}$$

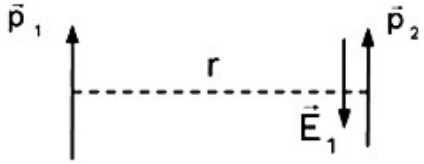
$\therefore$

$$\Rightarrow \quad \vec{F} = \frac{-6Kp_1p_2}{r^4}\hat{r}$$

Here –ve sign signifies that force is attractive in nature.
The same expression of force can also be calculated by vector addition of forces exerted on +ve and –ve charges of dipole $\vec{p}_2$ due to field $\vec{E}_1$, produced by $\vec{p}_1$.

(b) When dipoles are placed parallel to each other

The interaction energy between $\vec{p}_2$ and $\vec{E}_1$ is



$$U = -\vec{p}_2.\vec{E}_1 = -p_2\frac{Kp_1}{r^3}\cos 180^\circ \Rightarrow U = \frac{Kp_1p_2}{r^3}$$

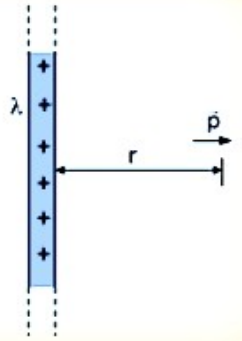
$$\because \quad \vec{F} = -\frac{dU}{dr}\hat{r} = -Kp_1p_2\frac{d}{dr}\left(\frac{1}{r^3}\right)\hat{r}$$

$$\vec{F} = \frac{3Kp_1p_2}{r^4}\hat{r}$$

(Force is repulsive in nature)

Note: If we calculate the net torque on dipole $\vec{p}_2$ in above both cases, it will be zero. It means a non-uniform field can give zero torque, but it always gives non-zero force.

Q. **Calculate the force acting on a dipole placed near a long charged wire as shown in figure.**



Sol: Electric field due to line charge near the dipole, $\vec{E} = \frac{2K\lambda}{r}\hat{r}$.

Interaction energy between the field and dipole is

$$U = -\vec{p}.\vec{E} \Rightarrow U = -\frac{2Kp\lambda}{r}\cos 0^\circ \Rightarrow U = -\frac{2Kp\lambda}{r}$$

$$\because \quad \vec{F}_{dipole} = -\frac{dU}{dr}\hat{r}$$

$$\therefore \quad \vec{F} = 2Kp\lambda\frac{d}{dr}\left(\frac{1}{r}\right)\hat{r}$$

$$\vec{F} = -\left(\frac{2Kp\lambda}{r^2}\right)\hat{r}$$ (Force is attractive in nature).